Babu Banarasi Das University

Case Study
ON

Job Attributes and Salary Dataset Analysis using Hadoop

Submitted To:-

Mr. Vikash (IBM)

Submitted By:-

Name – Chahat Singh

Section – BCADS23

Roll Number – 1240258153

PROJECT

Job Attributes and salary Dataset MapReduce Practical

- **Architecture:**

  - **The architecture of this project follows a standard Big Data pipeline using Hadoop ecosystem components.**

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- **Explanation:**

  - **Raw dataset (Job Attributes and salary data) is first stored in HDFS**
  - **MapReduce is used to process the data and calculate average salary**
  - **Processed data is stored back in HDFS**
  - **Hive is used to perform analytical queries**
  - **Final insights are generated from the results**

- **Tools Used:**

  - **Hadoop Distributed File System (HDFS)**
  - **MapReduce (Hadoop Streaming using Python)**
  - **Hive (for data analysis)**
  - **Linux Terminal**
  - **ChatGPT (Generative AI tool)**

- **Technologies:**

  - **Python (for mapper and reducer)**
  - **CSV dataset**
  - **Hadoop Ecosystem**

- **Use of Generative AI in the Project:**

  - **In this project, Generative AI tools were used to assist in development and understanding of Big Data concepts.**

- **Areas Where AI Was Used:**

  1. **Code Generation**

     - **Generative AI was used to generate MapReduce programs (mapper and reducer).**

  2. **Debugging**

     - **Errors in code were identified and fixed using AI assistance.**

  3. **Query Writing**

     - **Hive queries were generated and optimized using AI.**

  4. **Concept Understanding**

     - **AI helped in understanding:**
       - **HDFS architecture**
       - **MapReduce workflow**
       - **Hive operations**

- **Example Prompt Used:**

  - **Write a Python MapReduce program to calculate average job salary.**

- **Benefits of Using AI:**

  - **Faster development**
  - **Reduced coding errors**
  - **Better understanding of concepts**
  - **Improved efficiency**

- **Validation:**

  - **All AI-generated code was manually tested and verified before execution.**

- **Dataset Description:**

  - **The dataset used in this project is a Job Attributes and salary dataset containing user salary and job details. The data is stored in CSV (Comma Separated Values) format and is used for analysis using Hadoop.**

- **Dataset Structure:**

- **Job Title – Role of the employee**
- **Experience Years – Total work experience**
- **Education Level – Qualification (e.g., Bachelor's, Master's)**
- **Skills Count – Number of skills possessed**
- **Industry – Sector of work**
- **Company Size – Small, Medium, or Large organization**
- **Location – Job location**
- **Remote Work – Yes/No (work from home option)**
- **Certifications – Professional certifications count**
- **Salary – Compensation earned Dataset Source:**

  - **The dataset is a combination of different jobs data and custom entries created for project demonstration.**

- **Purpose of Dataset:**

  - **The dataset is used to:**
  - **Analyze job industry**
  - **Find good jobs**
  - **Perform data aggregation using MapReduce**
  - **Generate insights using Hive**

- **Sample Data:**

```
job_id,job_title,experience_years,education_level,skills_count,industry,company_
1,Software Engineer,2,Bachelor,5,IT,Medium,India,Yes,1,600000
2,Data Analyst,3,Master,6,Analytics,Large,USA,No,2,850000
3,Web Developer,1,Bachelor,4,IT,Small,India,Yes,0,400000
4,Project Manager,6,Master,8,Management,Large,UK,No,3,1200000
5,Machine Learning Engineer,4,Master,7,AI,Medium,USA,Yes,2,1500000
6,System Administrator,3,Bachelor,5,IT,Medium,Canada,No,1,700000
7,HR Executive,2,Bachelor,3,HR,Small,India,No,0,350000
8,Cloud Engineer,5,Master,6,Cloud,Large,USA,Yes,3,1400000
9,Business Analyst,4,Master,5,Finance,Medium,UK,Yes,2,900000
10,DevOps Engineer,3,Bachelor,6,IT,Large,India,Yes,2,1100000
```

➢ Conclusion:

- In this project, a complete Big Data pipeline was successfully implemented using Hadoop ecosystem components.
- The dataset was stored in HDFS, processed using MapReduce to calculate average job ratings, and analyzed using Hive to extract meaningful insights such as top-salary jobs.
- The project helped in understanding the working of distributed data processing and data analysis techniques.
- Generative AI acted as a smart assistant throughout the project by helping in code generation, debugging, and understanding complex concepts.

❖ Generative AI was used not only for code generation but also as a learning assistant to understand Big Data workflows and improve overall efficiency.

➢ Question-1:- job salary Analysis using Hadoop Streaming.

❖ Aim: - "To analyze job_salary dataset and calculate average rating using Hadoop MapReduce."

Step-01: - Create a “CSV” file.

- Cat > jobs_salary.csv

```
[cloudera@quickstart ~]$ cat > jobs_salary.csv
job_id,job_title,experience_years,education_level,skills_count,industry,company_
size,location,remote_work,certifications,salary
1,Software Engineer,2,Bachelor,5,IT,Medium,India,Yes,1,600000
2,Data Analyst,3,Master,6,Analytics,Large,USA,No,2,850000
3,Web Developer,1,Bachelor,4,IT,Small,India,Yes,0,400000
4,Project Manager,6,Master,8,Management,Large,UK,No,3,1200000
5,Machine Learning Engineer,4,Master,7,AI,Medium,USA,Yes,2,1500000
6,System Administrator,3,Bachelor,5,IT,Medium,Canada,No,1,700000
7,HR Executive,2,Bachelor,3,HR,Small,India,No,0,350000
8,Cloud Engineer,5,Master,6,Cloud,Large,USA,Yes,3,1400000
9,Business Analyst,4,Master,5,Finance,Medium,UK,Yes,2,900000
10,DevOps Engineer,3,Bachelor,6,IT,Large,India,Yes,2,1100000
11,Data Scientist,5,Master,8,AI,Large,USA,Yes,3,1600000
12,UI Designer,2,Bachelor,4,Design,Small,India,Yes,1,500000
13,Network Engineer,4,Bachelor,6,IT,Medium,Canada,No,2,800000
14,Product Manager,6,MBA,7,Management,Large,USA,No,3,1700000
15,QA Engineer,3,Bachelor,5,IT,Medium,India,Yes,1,650000
16,Cyber Security Analyst,4,Master,6,Security,Large,UK,No,2,1200000
17,Database Administrator,5,Bachelor,6,IT,Medium,USA,Yes,2,1300000
18,AI Researcher,6,PhD,9,AI,Large,USA,Yes,4,2000000
19,Technical Support,1,Bachelor,3,IT,Small,India,No,0,300000
20,Marketing Manager,5,MBA,7,Marketing,Large,UK,No,3,1250000
21,Sales Executive,2,Bachelor,4,Sales,Medium,India,No,1,450000
22,Frontend Developer,3,Bachelor,5,IT,Medium,USA,Yes,2,900000
23,Backend Developer,4,Bachelor,6,IT,Large,Canada,Yes,2,1100000
24,Full Stack Developer,5,Bachelor,7,IT,Large,USA,Yes,3,1400000
25,Data Engineer,4,Master,6,Data,Large,USA,Yes,2,1350000
26,Business Consultant,6,MBA,7,Consulting,Large,UK,No,3,1500000
27,Operations Manager,5,Bachelor,6,Operations,Medium,India,No,2,950000
28,Content Writer,2,Bachelor,4,Media,Small,India,Yes,1,400000
29,Game Developer,3,Bachelor,5,IT,Medium,USA,Yes,2,1000000
30,Mobile App Developer,4,Bachelor,6,IT,Large,India,Yes,2,1200000
```

Step-02:- Create a directory

- hdfs dfs -mkdir /user/cloudera/chahat_singh
- hdfs dfs -mkdir /user/cloudera/chahat_singh/job_salary

```
[cloudera@quickstart ~]$ hdfs dfs -mkdir /user/cloudera/chahat_singh
[cloudera@quickstart ~]$ hdfs dfs -mkdir /user/cloudera/chahat_singh/job_salary
[cloudera@quickstart ~]$ hdfs dfs -ls /user/cloudera/chahat_singh/job_salary
[cloudera@quickstart ~]$ hdfs dfs -ls /user/cloudera/chahat_singh
Found 1 items
drwxr-xr-x   - cloudera cloudera          0 2026-04-22 11:01 /user/cloudera/chahat_singh/job_salary
[cloudera@quickstart ~]$
```

Step-03:- Load the “CSV” on Cloudera’s Server

- hdfs dfs -put movie.csv /user/cloudera/chahatsingh/job
- hdfs dfs -ls /user/cloudera/chahatsingh/job

```
[chahat@quickstart ~]$ hdfs dfs -put job.csv /user/chahat/job
[chahat@quickstart ~]$ hdfs dfs -ls /user/chahat/job
Found 1 items
-rw-r--r--    1 chahat chahat       1455 2026-04-20 08:41 /user/chahat/job/job.csv
[chahat@quickstart ~]$
```

Step-04:- Create a mapper file using “nano”

- **nano mapper.py**

```
#!/usr/bin/env python
import sys
for line in sys.stdin:
if "userId" in line:
continue
data = line.strip().split(',')
job = data[2]
salary = float(data[4])
```

- **print("%s\t%s" % (job, salary))**

```
#!/usr/bin/env python
import sys

for line in sys.stdin:
    if "userId" in line:
        continue
    data = line.strip().split(',')
    job = data[2]
    salary = float(data[10])
    print("%s\t%s" % (job, salary))
```

Step-05:- Create a reducer file using "nano"

- **nano reducer.py**
  - #!/usr/bin/env python
  - import sys

  - current_job = None
  - total_salary = 0
  - count = 0

  - for line in sys.stdin:
  - job_title, salary = line.strip().split('\t')
  - salary = float(salary)

  - if current_job == job_title:
    - total_salary += salary
    - count += 1
  - else:
    - if current_job:
    - print("%s\t%s" % (current_job, total_salary / count))
    - current_job = job_title
    - total_salary = salary
    - count = 1

  - # Last job title
  - if current_job:
  - print("%s\t%s" % (current_job, total_salary / count))

```
#!/usr/bin/env python
import sys

current_job = None
total = 0
count = 0

for line in sys.stdin:
    job, salary = line.strip().split('\t')
    salary = float(salary)

    if current_job == job:
        total += salary
        count += 1
    else:
        if current_job:
            print("%s\%s" % (current_job, total/count))
        current_job = job
        total = salary
        count = 1

if current_job:
    print("%s\t%s" % (current_job, total/count))
```

Step-06:- Make executable

- **chmod +x mapper.py reducer.py**

```
[cloudera@quickstart mapreduce]$ chmod +x mapper.py reducer.py
[cloudera@quickstart mapreduce]$ ls
mapper.py  reducer.py
[cloudera@quickstart mapreduce]$ ls -l
total 8
-rwxrwxr-x 1 cloudera cloudera 220 Apr 20 08:46 mapper.py
-rwxrwxr-x 1 cloudera cloudera 482 Apr 20 08:49 reducer.py
[cloudera@quickstart mapreduce]$
```

Step-07:- Copy "jobs_salary.csv" file, from /home/cloudera to/home/cloudera/ mapreduce

- **cp /home/cloudera/jobs_salary.csv /home/cloudera/mapreduce/job_salary.csv**

```
[cloudera@quickstart mapreduce]$ cp /home/cloudera/job_salary.csv /home/cloudera/mapreduce/job_salary.csv
[cloudera@quickstart mapreduce]$ ls
mapper.py  job_salary.csv  reducer.py
[cloudera@quickstart mapreduce]$
```

Step-08:- Local test

- **cat job.csv | python mapper.py**
- **cat job.csv | python mapper.py | sort | python reducer.py**

```
[cloudera@quickstart mapreduce]$ cat job_salary.csv | python mapper.py
Software Engineer   90000
Data Scientist      95000
Product Manager     105000
Data Analyst        70000
DevOps Engineer     85000
QA Engineer         60000
UI/UX Designer      65000
Business Analyst    75000
HR Manager          80000
Marketing Manager   78000
Sales Executive     58000
Content Writer      50000
Graphic Designer    55000
System Administrator 65000
Database Administrator 92000
Software Engineer   88000
Data Scientist      98000
Product Manager     110000
Data Analyst        72000
DevOps Engineer     87000
QA Engineer         62000
UI/UX Designer      67000
Business Analyst    77000
HR Manager          82000
Marketing Manager   79000
Sales Executive     59000
Content Writer      52000
Graphic Designer    56000
System Administrator 65000
[cloudera@quickstart mapreduce]$ cat job_salary.csv | python mapper.py | sort | python reducer.py
Business Analyst     76000.0
Content Writer       51000.0
Data Analyst         71000.0
Data Scientist       96500.0
Database Administrator 92500.0
DevOps Engineer      86000.0
Graphic Designer     55500.0
HR Manager           81000.0
Marketing Manager    78500.0
Product Manager      107500.0
QA Engineer          61000.0
Sales Executive      58500.0
Software Engineer    89000.0
System Administrator 65500.0
UI/UX Designer       66000.0
[cloudera@quickstart mapreduce]$
```

Step-09:- Run Hadoop Job

- **hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar -files mapper.py,reducer.py -input /user/cloudera/chahatsingh/job/job_salary.csv -output /user/cloudera/chahatsingh/job/output -mapper "python mapper.py" -reducer "python reducer.py"**

```
[cloudera@quickstart mapreduce]$ hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar -files mapper.py,reduc
er.py -input /user/cloudera/chahatsingh/job/job salary.csv -output /user/cloudera/chahatsingh/job/output -mapper
"python mapper.py" -reducer "python reducer.py"
packageJobJar: [/] [/usr/jars/hadoop-streaming-2.6.0-cdh5.4.2.jar] /tmp/streamjob7283949069123864319.jar tmpDir=null
26/04/20 08:58:31 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/04/20 08:58:31 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/04/20 08:58:33 INFO mapred.FileInputFormat: Total input paths to process : 1
26/04/20 08:58:33 INFO mapreduce.JobSubmitter: number of splits:2
26/04/20 08:58:33 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1776698867558_0002
26/04/20 08:58:35 INFO impl.YarnClientImpl: Submitted application application_1776698867558_0002
26/04/20 08:58:35 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1776698867558_0002/
26/04/20 08:58:35 INFO mapreduce.Job: Running job: job_1776698867558_0002
26/04/20 08:58:51 INFO mapreduce.Job: Job job_1776698867558_0002 running in uber mode : false
26/04/20 08:58:51 INFO mapreduce.Job: map 0%  reduce 0%
26/04/20 08:59:02 INFO mapreduce.Job: map 50% reduce 0%
26/04/20 08:59:03 INFO mapreduce.Job: map 100% reduce 0%
26/04/20 08:59:15 INFO mapreduce.Job: map 100% reduce 100%
26/04/20 08:59:16 INFO mapreduce.Job: Job job_1776698867558_0002 completed successfully
26/04/20 08:59:16 INFO mapreduce.Job: Counters: 49
```

Step-10:- View Output

- **hdfs dfs -cat /user/cloudera/chahatsingh/job/output/part-00000**

```
[cloudera@quickstart mapreduce]$ hdfs dfs -ls /user/cloudera/chahatsingh/job/output
Found 2 items
-rw-r--r--   1 cloudera cloudera     0 2026-04-20 08:59 /user/cloudera/chahatsingh/job/output/_SUCCESS
-rw-r--r--   1 cloudera cloudera   219 2026-04-20 08:59 /user/cloudera/chahatsingh/job/output/part-00000

[cloudera@quickstart mapreduce]$ hdfs dfs -cat /user/cloudera/chahatsingh/job/output/part-00000
Business Analyst        76000.0
Content Writer          51000.0
Data Analyst            71000.0
Data Scientist          96500.0
Database Administrator  92500.0
DevOps Engineer         86000.0
Graphic Designer        55500.0
HR Manager              81000.0
Marketing Manager       78500.0
Product Manager         107500.0
QA Engineer             61000.0
Sales Executive         58500.0
Software Engineer       89000.0
System Administrator    65500.0
UI/UX Designer          66000.0
[cloudera@quickstart mapreduce]$
```

- **<u>Question-2:-</u> To perform data analysis on processed job dataset using Hive.**

  - **Aim: - "To analyze job data and find future jobs using Hive queries."**

Step-01:- Open Hive

- **hive**

```
[cloudera@quickstart mapreduce]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j.properties
WARNING: Hive CLI is deprecated and migration to Beeline is recommended.
hive>
```

Step-02:- Create table on hive

- **CREATE TABLE job_avg (job STRING,**
  avg_rating FLOAT
)
  ROW FORMAT DELIMITED
  FIELDS TERMINATED BY '\t';

```
hive> CREATE TABLE job_avg (
    >  job STRING,
    >  avg_salarg FLOAT
    > )
    > ROW FORMAT DELIMITED
    > FIELDS TERMINATED BY '\t';
OK
Time taken: 4.05 seconds
hive> show tables;
OK
count_string
job_avg
Time taken: 0.382 seconds, Fetched: 2 row(s)
hive>
```

Step-03:- Load data on the table in job_avg

- **LOAD DATA INPATH '/user/cloudera/chahatsingh/job/output' INTO TABLE job_avg;**

```
hive> LOAD DATA INPATH '/user/cloudera/chahatsingh/job/output' INTO TABLE job_avg;
Loading data to table default.job_avg
chgrp: changing ownership of 'hdfs://quickstart.cloudera:8020/user/hive/warehouse/job_avg/part-00000':
User does not belong to hive
Table default.job_avg stats: [numFiles=1, totalSize=219]
OK
Time taken: 1.471 seconds
hive>
```

Step-04:- View Data

Select * from job_avg;

```
hive> SELECT * FROM job_avg;
OK
Data Scientist          95000.00
Data Analyst            65000.00
DevOps Engineer         90000.00
Database Administrator  98000.00
Business Analyst        72000.00
QA Engineer             60000.00
HR Manager              75000.00
Marketing Manager       70000.00
Product Manager        120000.00
Sales Executive         55000.00
Software Engineer       85000.00
UI/UX Designer          65000.00
Content Writer          45000.00
Graphic Designer        50000.00
System Administrator    68000.00
Network Engineer        80000.00

Time taken: 0.857 seconds, Fetched: 16 row(s)
hive>
```

Step-05:- Find Top job

- **Select * from job_avg order by avg_salary desc;**

```
hive> SELECT * FROM job_avg ORDER BY avg_salary DESC;
Query ID = cloudera_20260420090909_04fba5c2-85cf-43ea-aea4-2b7756c7d776
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1776698867558_0002, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1776698867558_
0002/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1776698867558_0002
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-04-20 09:10:09,363 Stage-1 map = 0%,  reduce = 0%,
2026-04-20 09:10:20,037 Stage-1 map = 100%,  reduce = 0%,  Cumulative CPU 3.05 sec
2026-04-20 09:10:31,713 Stage-1 map = 100%,  reduce = 100%,  Cumulative CPU 6.39 sec
MapReduce Total cumulative CPU time: 6 seconds 390 msec
Ended Job = job_1776698867558_0002
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 6.39 sec   HDFS Read: 5822 HDFS Write: 207 SUCCESS
Total MapReduce CPU Time Spent: 6 seconds 390 msec
OK
Data Scientist           95000.00
DevOps Engineer          90000.00
Database Administrator   98000.00
Product Manager         120000.00
Software Engineer        85000.00
Network Engineer         80000.00
HR Manager               75000.00
Business Analyst         72000.00
System Administrator     68000.00
Data Analyst             65000.00
UI/UX Designer           65000.00
QA Engineer              60000.00
Sales Executive          55000.00
Graphic Designer         50000.00
Content Writer           45000.00
Marketing Manager        70000.00

Time taken: 40.217 seconds, Fetched: 16 row(s)
hive>
```

Step-06:- Top 5 job

Select * from job_avg order by avg_salary desc limit 5;

```
hive> SELECT * FROM job_avg ORDER BY avg_salary DESC LIMIT 5;
Query ID = cloudera_20260420091212_47c4e033-f57b-415a-9f23-cca2fe7d76d5
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1776698867558_0003, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1776698867558_0003/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1776698867558_0003
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-04-20 09:13:05,391 Stage-1 map = 0%,   reduce = 0%,
2026-04-20 09:13:14,846 Stage-1 map = 100%,  reduce = 0%,  Cumulative CPU 2.05 sec
2026-04-20 09:13:26,308 Stage-1 map = 100%,  reduce = 100%,  Cumulative CPU 5.01 sec
MapReduce Total cumulative CPU time: 5 seconds 10 msec
Ended Job = job_1776698867558_0003
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 5.01 sec    HDFS Read: 5961 HDFS Write: 57 SUCCESS
Total MapReduce CPU Time Spent: 5 seconds 10 msec
OK
job                    avg_salary
------------------     -------------
Product Manager         210000.00
Data Scientist          185000.00
Database Administrator  170000.00
DevOps Engineer         160000.00
Software Engineer       145000.00

Time taken: 35.365 seconds, Fetched: 5 row(s)
hive>
```

- **Output:**

  - **Hive successfully displayed:**
  - **Average ratings**
  - **Top-salary job**

✞ **In this practical, Hive was used to analyze processed job data and retrieve meaningful insights using SQL-like queries.**